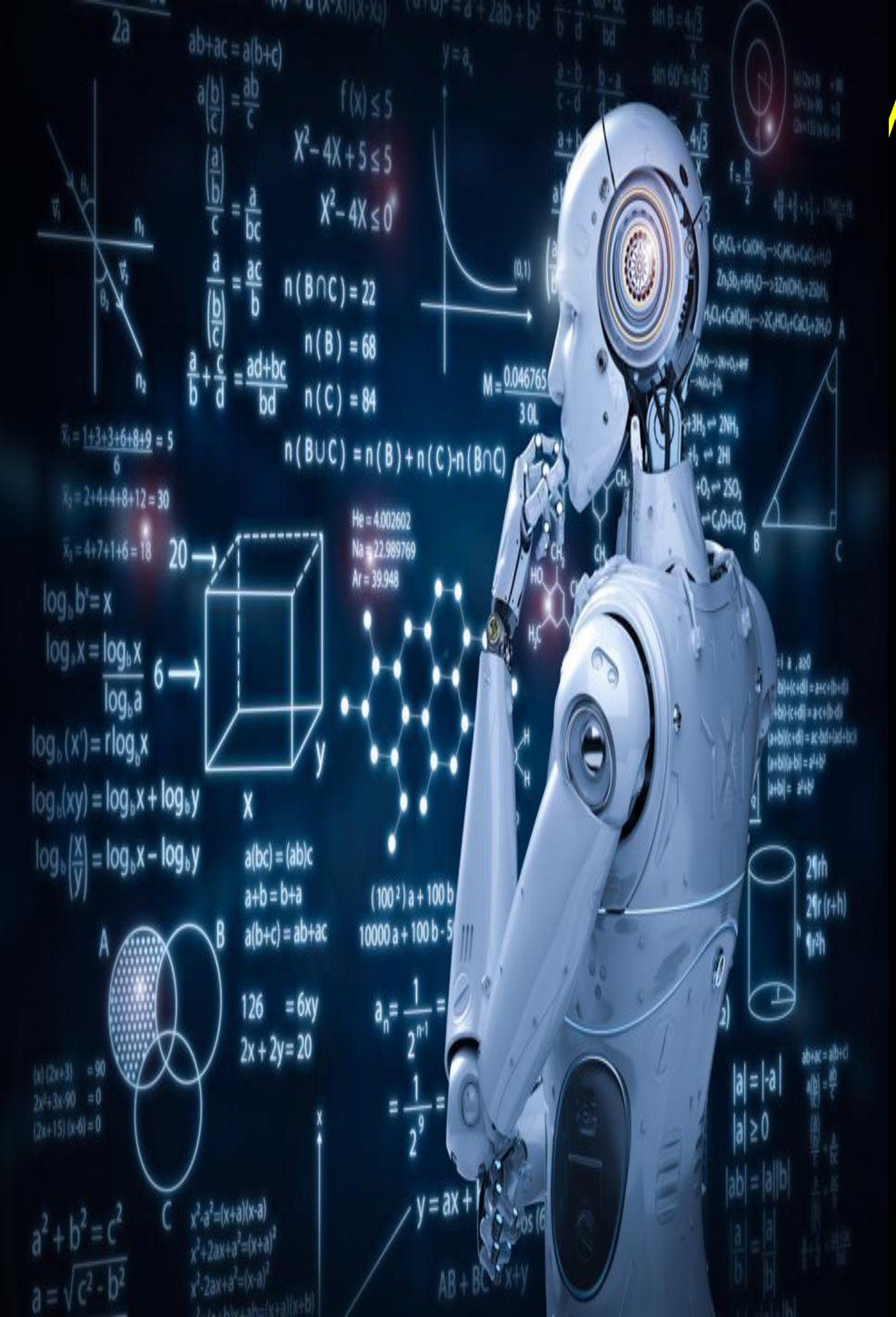
PGCP-AI

PRACTICAL MACHINE LEARNING

Day 3: Data Pre-processing & Feature Engineering

Dr. Kiran Waghmare
CDAC Mumbai



Agenda

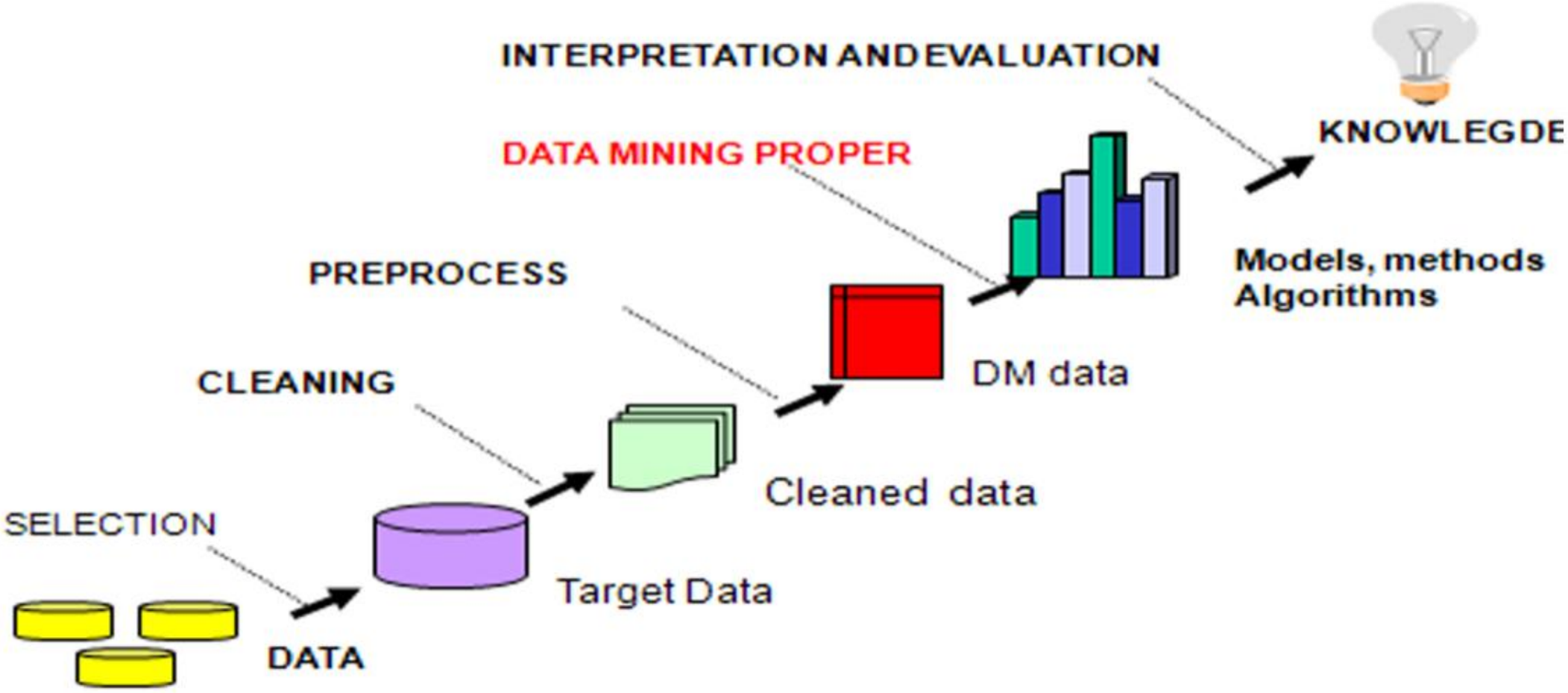
- Visualization
- Feature Engineering
- Pre-processing
- Transformations
- Feature scaling

KDD

Knowledge Discovery Process

- Goal
  - understanding the application domain, and goals of KDD effort
- Data selection, acquisition, integration
- Data cleaning
  - noise, missing data, outliers,etc.
- Exploratory data analysis
  - dimensionality reduction, transformations
  - selection of appropriate model for analysis, hypotheses to test
- Data mining
  - selecting appropriate method that match set goals (classification, regression, clustering, etc)
  - selecting algorithm
- Testing and verification
- Interpretation
- Consolidation and use

DM- KDD process
(re-iterated if needed)



Steps of a KDD Process

- Learning the application domain:
  - relevant prior knowledge and goals of application
- Creating a target data set: data selection
- Data cleaning and preprocessing: (may take 60% of effort!)
- Data reduction and transformation:
  - Find useful features, dimensionality/variable reduction, invariant representation.
- Choosing functions of data mining
  - summarization, classification, regression, association, clustering.
- Choosing the mining algorithm(s)
- Data mining: search for patterns of interest
- Pattern evaluation and knowledge presentation
  - visualization, transformation, removing redundant patterns, etc.
- Use of discovered knowledge

Machine Learning Operations

Feature Engineering

Feature engineering is the process of using domain knowledge and data transformation techniques to create input variables (features) that make machine learning models more effective.

In simple terms:

It's about preparing and transforming raw data into a format that helps a machine learning model learn better.

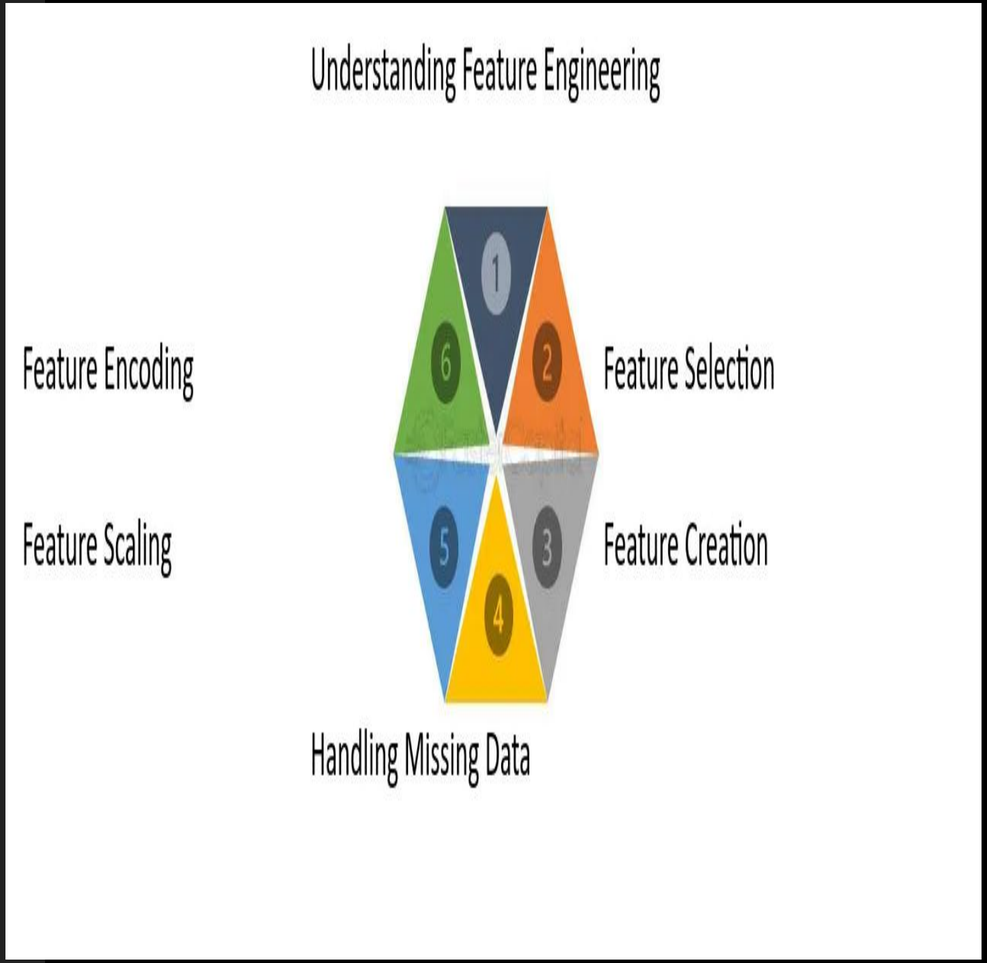
Why it's important:

Good features = better model performance.

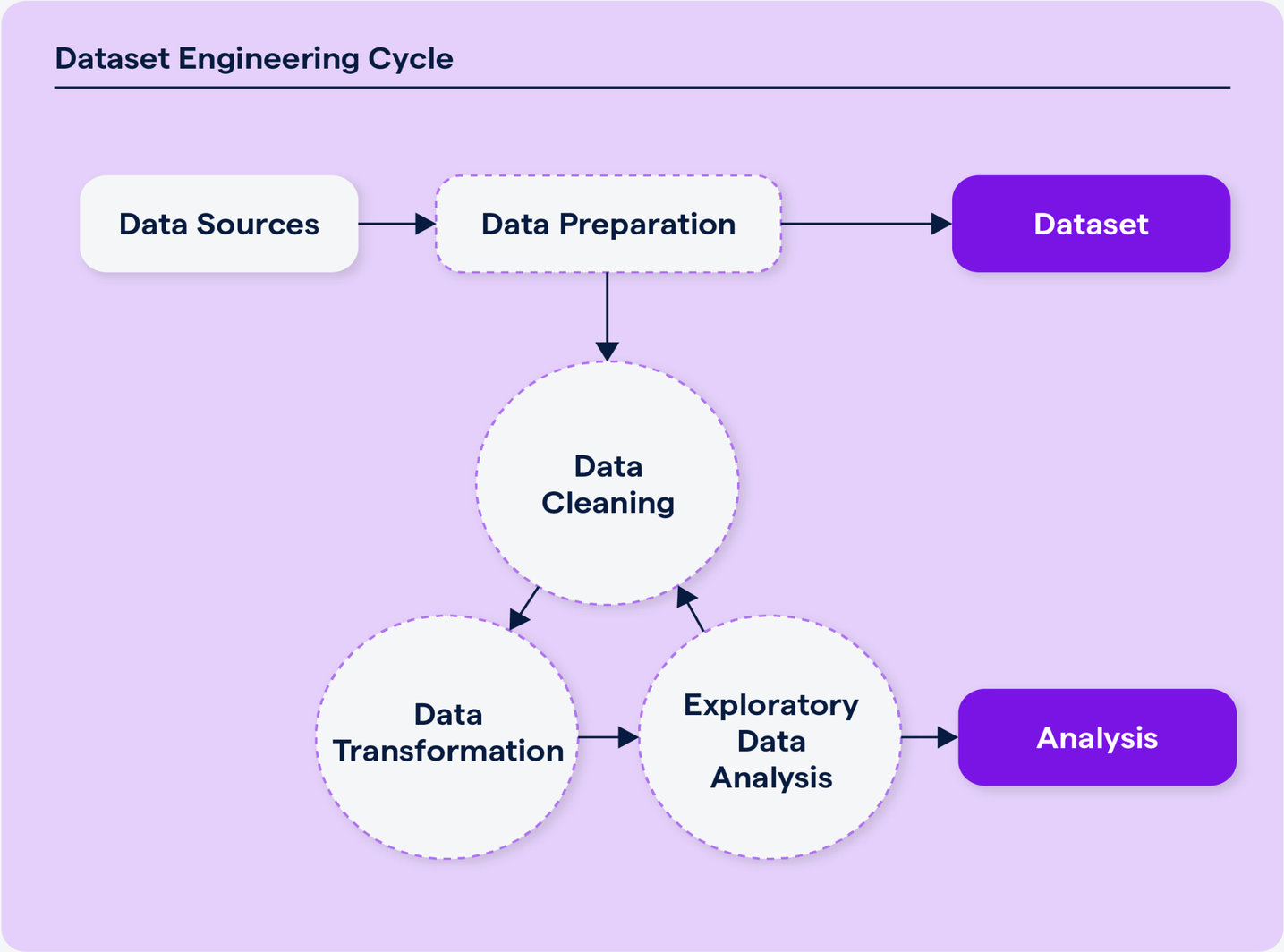
A machine learning model can only learn from the data it's given. If the features are poorly designed or irrelevant, the model will struggle.

Common feature engineering tasks:

1. **Cleaning data** – handling missing values, fixing errors, etc.
2. **Encoding categorical variables** – e.g., converting "red", "blue", "green" to numbers.
3. **Scaling/normalizing** – adjusting values so they're on similar scales.
4. **Creating new features** – e.g., turning a timestamp into day of the week or hour of the day.
5. **Combining features** – e.g., multiplying or summing two features to create a new one.
6. **Reducing dimensionality** – e.g., using PCA to reduce redundant or correlated features.



The Model Development Phase



Feature Engineering Cycle

Model Training & Evaluation Cycle

Feature engineering methods

Data pre-processing

- Aggregation
- Sampling
- Discretization and binarization
- Attribute transformation
- Feature creation
- Feature selection
  - Choose subset of existing features

Converting to Numerical Features

- Often want a real-valued example representation:

Age	City	Income
23	Van	22,000.00
23	Bur	21,000.00
22	Van	0.00
25	Sur	57,000.00
19	Bur	13,500.00
22	Van	20,000.00

→

Age	Van	Bur	Sur	Income
23	1	0	0	22,000.00
23	0	1	0	21,000.00
22	1	0	0	0.00
25	0	0	1	57,000.00
19	0	1	0	13,500.00
22	1	0	0	20,000.00

- This is called a "1 of k" encoding.
- We can now interpret examples as points in space:
  - E.g., first example is at (23,1,0,0,22000).

Feature Aggregation

- Feature aggregation:
  - Combine features to form new features:

Van	Bur	Sur	Edm	Cal
1	0	0	0	0
0	1	0	0	0
1	0	0	0	0
0	0	0	1	0
0	0	0	0	1
0	0	1	0	0

→

BC	AB
1	0
1	0
1	0
0	1
0	1
1	0

- Fewer province “coupons” to collect than city “coupons”.

Feature Selection

- Feature Selection:
  - Remove features that are not relevant to the task.

SID:	Age	Job?	City	Rating	Income
3457	23	Yes	Van	A	22,000.00
1247	23	Yes	Bur	BBB	21,000.00
6421	22	No	Van	CC	0.00
1235	25	Yes	Sur	AAA	57,000.00
8976	19	No	Bur	BB	13,500.00
2345	22	Yes	Van	A	20,000.00

  - Student ID is probably not relevant.

Feature Transformation

- Mathematical transformations:
  - Discretization (binning): turn numerical data into categorical.

Age
23
23
22
25
19
22

→

< 20	>= 20, < 25	>= 25
0	1	0
0	1	0
0	1	0
0	0	1
1	0	0
0	1	0

- Only need consider 3 values.

Feature Transformation

- Mathematical transformations:
  - Discretization (binning): turn numerical data into categorical.
  - Square, exponentiation, logarithm, and so on.

1.00

EDA

explore the data

to analyze

examining the data in detail
understand and interpret it

summarize it

main characteristics of the data

Constant Learners

Mute (m)

1:35 / 13:40 • definition of exploratory data analysis

EDA

understand the data
consists of
variables - relationship

useful for us or
not

identify errors
discover patterns

best way to manipulate
the datasets

&

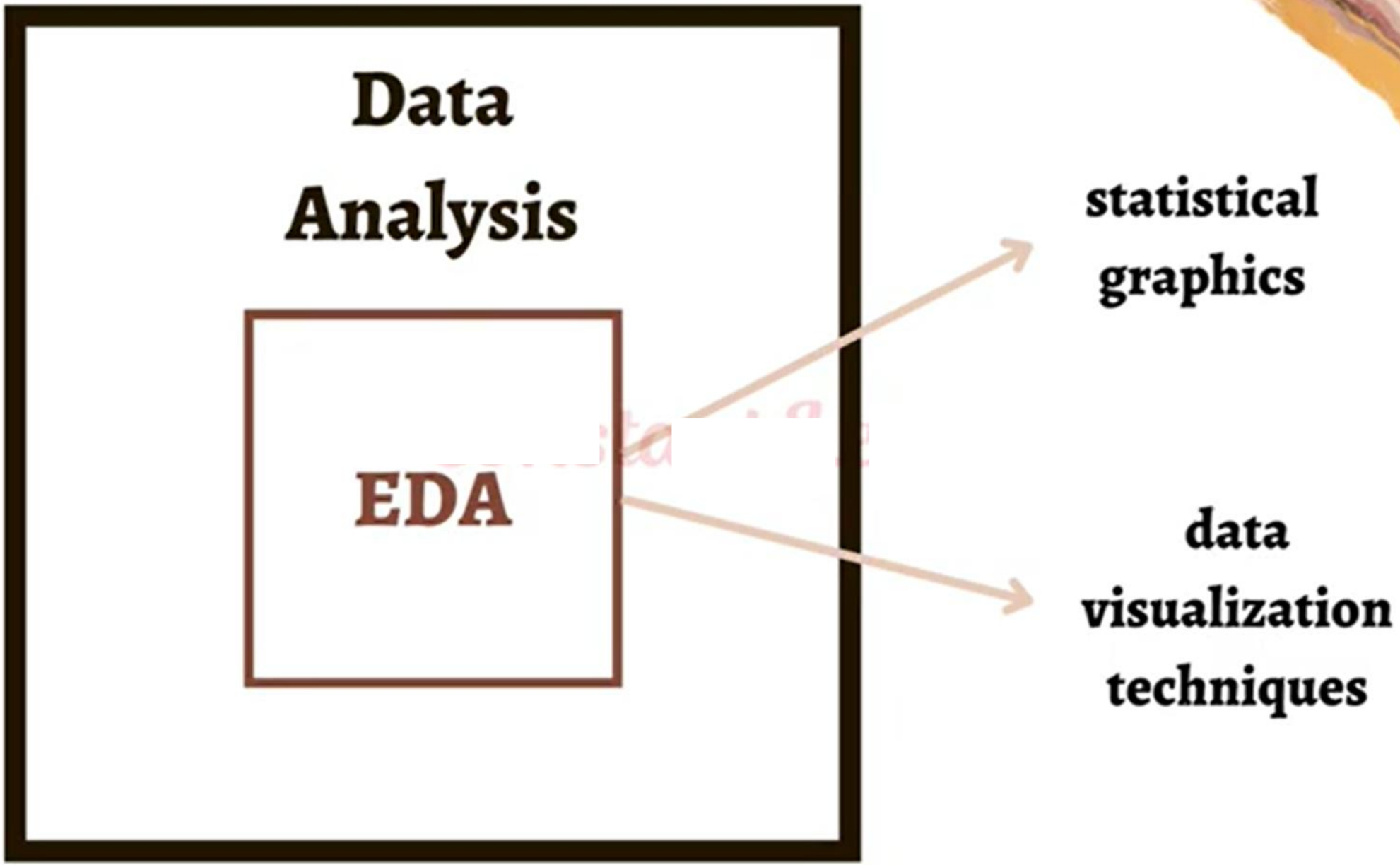
spot anomalies
detect the outliers

formulate hypotheses

initial
explanation

new data collection
& experiments

Constant Learners


Data
Analysis
EDA
statistical
graphics
data
visualization
techniques

EDA

- **Univariate**
  - Graphical
  - Non-graphical
- **Multivariate**
  - Graphical
  - Non-graphical

Exploratory Data Analysis

- You should always ‘look’ at the data first.

- But how do you ‘look’ at features and high-dimensional examples?
  - Summary statistics.
  - Visualization.
  - ML + DM (later in course).

Interview Questions : Data Cleaning & Outliers

- What are outliers?
- How do you detect outliers (Z-score, IQR)?
- How do you handle outliers?
- What is data smoothing?
- What is binning?
- What is noise reduction?
- What is duplicate data and how do you handle it?
- What is data consistency?
- What is data validation?
- What tools do you use for data cleaning?

Interview Questions : Feature Scaling & Transformation

- What is feature scaling?
- Why is feature scaling necessary?
- What is normalization?
- What is standardization?
- Difference between normalization and standardization?
- What is Min-Max scaling?
- What is Z-score scaling?
- When should you scale data?
- Which algorithms require feature scaling?
- What happens if you don’t scale data?

Interview Questions : Encoding Categorical Data & Advanced

- What is categorical data?
- What is label encoding?
- What is one-hot encoding?
- What are the drawbacks of one-hot encoding?
- When would you use label encoding vs one-hot encoding?
- What is data leakage?
- How do you prevent data leakage during preprocessing?
- What is a preprocessing pipeline?
- Why should preprocessing be done after train-test split?
- What is feature engineering with examples?
- How do you handle imbalanced datasets?
- How do you preprocess text data in NLP tasks?

THANK YOU